

Helix OTA — Android Agent Build Integration (1.0.0-MVP)

Field	Value
Revision	1
Created	2026-06-07
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Status	active
Status summary	How the Helix OTA Android agent (ota-android-agent) is built and shipped as a system app in the AOSP tree for the Orange Pi 5 Max / RK3588 target, so it can use the @SystemApi UpdateEngine and run as android.uid.system (or as a privileged system app). Covers the Android.bp system-module build, platform signing, sharedUserId / priv-app placement, SELinux + /data/ota_package/ access, and how the OTA package itself is generated/signed by the Helix Go build pipeline (ota_from_target_files, ZIP_STORED, --package_key / --payload_signer). Whether the chosen RK3588 Android-15 BSP ships AOSP A/B + dynamic partitions + Virtual A/B + dm-user (vs an A-only Rockchip proprietary OTA) is UNVERIFIED and is the top-priority hardware gate. The exact SELinux policy to let a Helix-signed system app reach update_engine, and the precise Android.bp certificate:/privileged: fields for the target, are UNVERIFIED. HelixConstitution clause numbers are UNVERIFIED.
Issues	Whether the chosen RK3588 Android-15 BSP ships AOSP A/B + dynamic partitions + Virtual A/B + dm-user (vs an A-only Rockchip proprietary OTA) is UNVERIFIED and is the top-priority hardware gate. The exact SELinux policy to let a Helix-signed system app reach update_engine, and the precise Android.bp certificate:/privileged: fields for the target, are UNVERIFIED. HelixConstitution clause numbers are UNVERIFIED.
Fixed	N/A (initial revision).
Continuation	Validate the target image is AOSP A/B + VAB on real hardware (apply a test OTA, observe MergeStatus, force failed boot + power-loss-mid-merge); pin the Android.bp certificate/priv-

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	app config and SELinux .te policy; integrate ota_from_target_files with ZIP_STORED + --package_key (otacert-pinned) + --payload_signer into the Go pipeline; decide HSM-backed payload signing + key rotation.

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ToC mandated by HelixConstitution §11.4.61 (UNVERIFIED clause number).

1. Purpose and scope

This spec covers **building and shipping** the Helix OTA Android agent as a **system component** for the 1.0.0-MVP target (Orange Pi 5 Max / RK3588, Android 15). It also covers the build/signing of the **OTA package** the agent consumes, since the Helix Go build pipeline owns that contract.

It honors the LOCKED strategy — native Android A/B (update_engine + AVB/dm-verity + auto-rollback) [ADR-0001 §1] — and the LOCKED MVP trust model — signing + SHA-256 + AVB, TUF device-side deferred to 1.0.1 [ADR-0002 §4.1, §4.3]. The runtime behaviour of the agent is in [integration_guide.md](#); the apply contract is in [update_engine_integration.md](#).

2. Why a system app (recap)

android.os.UpdateEngine is @SystemApi with **no third-party manifest permission**; only system apps can call it [android-update-engine-api §10]. The agent therefore must be built into the system image as either a **system-UID app** or a **privileged system app**, with access to /data/ota_package/ and SELinux permission to bind update_engine [android-update-engine-api §10]. It cannot be a Play-Store app.

3. Building the agent in the AOSP tree (Android.bp)

The agent module is built **inside the AOSP tree** (Soong/Android.bp) so it can link the hidden/@SystemApi UpdateEngine surface and be platform-signed. A representative Android.bp is in [code_snippets/Android.bp](#). Key points:

- `platform_apis`: `true` (or an appropriate `sdk_version/system_current`) so the build resolves `android.os.UpdateEngine / UpdateEngineCallback`.
- `certificate`: "platform" for the platform-signing path (§5), or a Helix-controlled certificate registered in the device's `mac_permissions/keys`.
- `privileged`: `true` **and** placement under `priv-app` if using the privileged-app path (§4); omit if using system-UID.
- The KMP `ota-android-agent` Kotlin output (`androidMain + commonMain`) is compiled to the module; the `ota-update-engine-bridge` is linked as the Android-only apply path.

The KMP layers (Auth-KMP, Security-KMP, Storage-KMP, Config-KMP) are consumed as the agent's dependencies; only canonical catalogue names are referenced (§7.1 anti-bluff; [documentation_standards.md §9](#)).

4. System UID vs privileged-app placement

Two valid deployment shapes [android-update-engine-api §10]:

Path	How	Trade-off
System UID	<code>android:sharedUserId="android.uid.system"</code> in the manifest and platform-signed	Strongest access (runs as the system user); broadest privilege — use only if the agent genuinely needs it. See code_snippets/AndroidManifest.xml .
Privileged system app	Install under <code>/system/priv-app/</code> (or <code>/vendor/priv-app/</code>), platform/OEM-signed, with a <code>priv-app</code> allowlist entry	Narrower than full system UID; still grants @SystemApi access via signature + <code>priv-app</code> status.

MVP picks the path the target image supports; both require platform/OEM signing and system-image placement. The choice is recorded per board and is UNVERIFIED until confirmed on the RK3588 image (§7).

5. Platform signing

OTA packages and the agent both depend on signing. For the **agent**: platform-signing (or a Helix-registered system certificate) is what unlocks @SystemApi [android-update-engine-api §10]. For the **OTA package**, two distinct signatures apply [android-update-engine-api §11]:

1. **Whole-package signature** — `-k / --package_key` (default `default_system_dev_certificate` from `META/misc_info.txt`, else `build/target/product/security/testkey`); performed by `signapk`; verified on device against `otacert`.
2. **A/B payload + metadata signature** — `--payload_signer` (default: `openssl pkeyutl` with the package private key); checked by `update_engine` via `METADATA_HASH / payload pubkey` verification.

Each key is a `.x509.pem` cert + a `.pk8` private key; the `.pk8` is kept secret [android-update-engine-api §11]. Helix must sign the package with a **Helix-controlled --package_key** whose cert is embedded as `otacert` and **pinned on device**, and sign the payload via `--payload_signer`. HSM-backed `--payload_signer` and key rotation are 1.0.1 follow-ups (Continuation).

6. SELinux and /data/ota_package access

The agent's SELinux domain must be allowed to:

- bind/talk to the `update_engine` service over Binder/AIDL [android-update-engine-api §10];
- read/write **/data/ota_package/** for the local `file://` apply path [android-update-engine-api §7, §10].

The exact `.te` policy for a Helix-signed system app reaching `update_engine` is **UNVERIFIED** and must be authored against the target tree (Continuation). The agent **must not** use SELinux escalation to bypass any AVB/dm-verity protection [android-avb-rollback §10].

7. Target board: Orange Pi 5 Max / RK3588

Top-priority gate — **UNVERIFIED**. AOSP documents the A/B + Virtual A/B *mechanism*; it does not certify the RK3588 / Orange Pi 5 Max. Whether the chosen Android-15 image ships the native path depends entirely on the **Rockchip BSP and the integrator's board config** [android15-virtual-ab §11]. Before the no-corruption guarantee can be claimed, validate on real hardware:

1. **Is the build AOSP A/B (not A-only)?** Many Rockchip images historically ship **A-only with a proprietary OTA** (`rkupdate / update.img`) rather than `update_engine` A/B. If A-only, the native path does not apply and a separate ADR is needed [android15-virtual-ab §11].
2. **Dynamic partitions enabled?** Confirm `super / logical` partitions (`BOARD_SUPER_PARTITION_SIZE`, `update groups`) [android15-virtual-ab §3, §11].

3. **VAB + VABc enabled?** Confirm `PRODUCT_VIRTUAL_AB_OTA / vabc_features.mk / PRODUCT_VIRTUAL_AB_COMPRESSION_METHOD` (default `lz4`) [android15-virtual-ab §7, §11].
4. **Kernel support.** RK3588 vendor kernels are typically 5.10-class; VAB needs `CONFIG_DM_SNAPSHOT`, compressed VAB needs `CONFIG_DM_USER` (the non-upstream `dm-user` module must be carried in the Rockchip kernel; load in first-stage ramdisk if modular) [android15-virtual-ab §6, §11].
5. **super sizing + storage.** Size super per [android15-virtual-ab §8]; prefer `lz4` to limit COW spill into `/data`; eMMC/NVMe endurance affects merge time — benchmark.
6. **Bootloader / AVB.** RK3588 uses U-Boot + Rockchip miniloader; confirm `boot_control` is the AOSP HAL (not a Rockchip stub), AVB/vbmeta signed, and rollback-index storage present [android-avb-rollback §12; android15-virtual-ab §11].

The validation procedure: apply a test OTA, observe `MergeStatus` transitions, force a failed boot to confirm automatic rollback, and force power loss mid-merge to confirm resume [android15-virtual-ab §11].

The compression knob (default `lz4`, `zstd` per fleet segment) is set at build time to match the device build; the agent's payload must target the device's enabled compression [android15-virtual-ab §12]. See [code_snippets/device_vab.mk](#).

8. OTA package generation and signing (Go pipeline)

The Helix **Go build pipeline** owns OTA-package production via `ota_from_target_files` [android-update-engine-api §11]:

- **Full** OTA from one `target-files.zip`; **incremental/delta** by diffing against a previous `target-files.zip` (delta deferred per [ADR-0005](#)).
- Produce the package with **ZIP_STORED** entries and per-entry offset/size streaming metadata (so the streaming opt-in works and the agent can compute the `payload.bin` offset/size) [android-update-engine-api §9].
- Emit `payload.bin` + `payload_properties.txt` (the four hash/size values), and surface those four values into the **control-plane release manifest** so the agent can pre-verify and pass them to `applyPayload` [android-update-engine-api §4, §11].
- Sign with the Helix `--package_key` (otacert-pinned) and `--payload_signer` (§5).
- Compute the **artifact-level SHA-256 + detached signature** that the agent verifies before apply (the MVP “signing + SHA-256” layer) [ADR-0002 §4.1]; this is in addition to the two AOSP package signatures.

This pipeline runs on the containers substrate (§11.4.76, UNVERIFIED) and conforms to the LOCKED stack — Go + Gin + REST `/api/v1`; HTTP/3 (QUIC) via `vasic-digital/http3` with HTTP/2 + `gzip` fallback; Brotli for control-plane responses, while the **artifact path stays identity + Range** [ADR-0004 §97, §106]. Representative invocation: [code_snippets/build_ota_package.sh](#).

9. Testing (four-layer)

Per HelixConstitution §1 / §1.1 and master design §13 (no-bluff positive evidence, §7.1):

1. **Source-presence gate.** Assert the `Android.bp` declares `platform_apis/` system signing and (system-UID path) the manifest carries `sharedUserId="android.uid.system"`; assert the pipeline invokes `ota_from_target_files` with `ZIP_STORED`, `--package_key`, and `--payload_signer`. Grep gates in CI.
2. **Artifact gate (bytes shipped).** Inspect the built OTA `.zip`: entries are `ZIP_STORED`; `payload.bin`, `payload_properties.txt`, and `META-INF/.../otacert` are present; verify the package signature and the four `payload_properties` values against the manifest. Inspect the built agent module: `platform-signed`, placed in `system/priv-app`.
3. **Runtime / integration.** On an emulator/host, install the system-signed agent and apply a generated `ZIP_STORED` package end-to-end. On the **real Orange Pi 5 Max**: the §7 validation procedure (apply → reboot → verify; corrupt-slot → confirm A/B fallback; power-loss-mid-merge → confirm resume) [master design §13; `android15-virtual-ab` §11].
4. **Mutation meta-test.** Negate a build invariant (e.g. drop `--payload_signer`, or build the package `ZIP_DEFLATED`, or strip the `otacert`, or remove `platform_apis`) and assert the gates flip `PASS` → `FAIL`. A build that ships an unsigned/mis-built package while gates stay green is a conformance defect.

10. Open / UNVERIFIED items

- RK3588 / Orange Pi 5 Max AOSP A/B + dynamic partitions + VAB + VABc + dm-user enablement — UNVERIFIED, top-priority [`android15-virtual-ab` §11, §13].
- Exact `Android.bp` certificate: `/privileged: config` and `priv-app allowlist` for the target — UNVERIFIED.
- SELinux `.te` policy for the Helix-signed agent to reach `update_engine` — UNVERIFIED [`android-update-engine-api` §13].
- HSM-backed `--payload_signer` and `vbmeta/package` key rotation — UNVERIFIED (1.0.1 follow-up) [`android-update-engine-api` §13].
- containers substrate (§11.4.76) — UNVERIFIED clause/binding.

11. Sources

- [android-update-engine-api.md](#) — `ota_from_target_files`, `ZIP_STORED`, `--package_key/--payload_signer`, system-app requirement.
- [android-avb-rollback.md](#), [android15-virtual-ab.md](#) — board/VAB considerations.
- [adr-0001-wrapped-engine.md](#), [adr-0002-supply-chain-trust.md](#), [adr-0004-transport.md](#), [adr-0005-delta-updates.md](#).
- [submodule_reuse_map.md](#), [documentation_standards.md](#), [2026-06-07-helix-ota-design.md](#).